

TriRAG Test Document

1. Retrieval and Reranking

Retrieval-Augmented Generation (RAG) combines document retrieval with language generation. A retriever first finds candidate chunks that may contain useful information for a question. A cross-encoder reranker is then applied to these candidates because it can compare the query and each candidate passage together and produce a more precise relevance score.

2. Why Reranking Happens After Retrieval

Initial retrieval is designed to be fast and scalable. Vector search can compare a query embedding against a large collection of document embeddings efficiently. However, embedding similarity is only an approximate measure of relevance. A cross-encoder performs a deeper query-passage interaction, so it is more computationally expensive. For this reason, systems normally retrieve a small candidate set first and rerank only those candidates.

3. Hybrid Retrieval

A hybrid RAG system can combine several retrieval methods. Dense retrieval uses embeddings to identify semantically similar passages. BM25 uses lexical matching and is useful when specific words, names, identifiers, or technical terms matter. Graph retrieval can follow relationships between entities and return related source chunks. Combining these methods can improve recall compared with relying on only one retriever.

4. Reciprocal Rank Fusion

Reciprocal Rank Fusion (RRF) combines ranked results from multiple retrievers. Instead of depending on incompatible raw relevance scores, RRF uses the rank position of each result. A chunk that appears near the top in several retrieval lists receives a stronger combined ranking. The fused list can then be passed to a reranker for more detailed relevance scoring.

5. Corrective RAG

Corrective RAG evaluates whether the retrieved evidence is good enough to answer a question. If the evidence is relevant, the system generates an answer using the retrieved context. If the evidence is weak, the system can rewrite or refine the query and perform retrieval again. A retry limit prevents an endless correction loop.

6. Example

Suppose the question is: Why is cross-encoder reranking performed after initial retrieval? A dense retriever may return passages about embeddings and ranking. BM25 may find passages containing the exact phrase cross-encoder. Graph retrieval may find passages connected to reranking and candidate selection. RRF combines these results, and the cross-encoder then selects the passages that most directly answer the question.